# Research Assignment 1.

## Section A: Data Fundamentals.

1. A database is a system used to store and organise data so that it can be easily accessed and managed. It helps people and businesses keep information safe and structured.
Examples: 1. Bank systems storing customer accounts
2. School systems storing student records,

2. A data warehouse is a system used to store large amount of historical data from different sources for analysis. It differs from regular database because with database it requires daily operation, it captures data via OLTP. Datawarehouse requires analysis and reporting it goes through OLAP. It is used to help businesses make better decisions using past data.

3. A data lake stores raw data in its original format (Structured and unstructured). Data warehouse is organised data, Data lake is raw data. Used when companies want to store large amounts of different types of data.

4. A data mart is a smaller part of a data warehouse focused on a specific department. Used by sales teams, HR, marketing teams.

Section B

11. AI is the ability of machines to perform tasks that require human intelligence
> Narrow AI - specific tasks (eg. voice assistants)
> General AI - human-like intelligence (not fully developed

12. Machine Learning allows machine to learn from data without being programmed.
Types
> Supervised -> uses labeled data
> Unsupervised -> finds patterns.
> Reinforcement -> Learns by reward.

13. Deep learning is a type of machine learning using neural networks.
Neural networks: systems that mimic the human brain to process data.

14. NLP allows machines to understand human language
Examples
> Chatbots
> Translation apps
> Voice assistants

15. LLMs are AI models trained on large text data to understand language.
Examples
> Chat GPT -> conversations
> Claude -> writing and analysis

[illegible] machine to learn [illegible]
[illegible]

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16. Generative AI creates new content like text, images or music

Tools

- Chat GPT -> text
- DALL. E -> images
- Midjourney -> art.

17. A training dataset is data used to teach AI models. Important because good data equals accurate results. Risk is that bad data causes bias and errors.

18. Computer vision allows machines to interpret images

Examples

- face recognition
- Self-driving cars.

Section C.

19. API allows different software systems to communicate. Example: Like a waiter taking orders between you and the kitchen

20. A code used to identify and secure access to an API. Risk if exposed: Hackers can misuse it.

21. REST = Representational State Transfer

Methods

- GET - retrieve data

> Post -> send data
> PUT -> update data
> DELETE -> remove data.

22. JSON is a format for storing and exchanging data. Example

JSON

```
{
  "name": "John",
  "age": 25
}
```

23. Promp engineering is the skill of writing clear and specific instructions (called prompts) to guide AI tools so they give accurate and useful responses
Important for improving the quality of AI responses
Examples : Bad: "Tell me about AI"
: Good: "Explain AI in simple terms with examples"

24. MAP fuction - applies a function to each item in a list.
Example (Python)
Python

```
numbers = [1, 2, 3]
result = list (map(lambda x: x* 2, numbers))
```

25. Cloud computing - delivering services over the internet.

- IaaS - infrastructure (AWS)
- PaaS - platform (Google App Engine)
- SaaS - Software (Gmail)

26. Version control - tracks changes in code.
Git: Tool for managing code versions.
- Commit - save changes
- Branch - separate versions

27. Jupyter Notebook - A tool used for coding and analysis
Language: Mostly Python
Why popular: Easy to use and visualize data.

28 Python is a programming language used in data and AI.
Libraries
- Pandas → data analysis
- NumPy - calculations
- Matploitlib - graphs
- Scikit-learn - ML

<u>Section D</u>.

29. Data privacy - Protecting personal data.
GDPR: a law in Europe that protects data privacy

30 Data ethics - using data responsibly
Examples

> Selling personal data
> Biased AI decisions
> Misusing data.

31. Data Governance - managing data properly.
Components:
> Security
> Quality
> Policies

32. Bias in AI -> When AI produces unfair results
Example: Hiring systems favoring certain groups.

33. Roles comparison.

| Roles | Work | Skills |
|---|---|---|
| Data Analyst | Analyze data. | Excel, SQL |
| Data Scientist | Build models | Python, ML |
| Data Engineer | Build system | Databases. |
| AI Engineer | Develop AI | ML, Deep Learning |

34. Business Intelligence uses data to support decisions
Tools:
> Power BI
> Tableau

## Section E

35. I found AI and machine learning interesting because they show how machines can learn and make decisions like humans.

36. Data modelling was difficult at first. I used online resources such as Chat GPT and watched videos from Youtube to understant it better.

37. For future impact, data and AI will help me in my career by improving decision making and creating job opportunities. For example, businesses use data to understand customers better.

References


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